

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR
Master of Science in Computer Engineering
TECHNICAL SKILLS

Aug 2024 – May 2026

Programming Languages: Python, C++, C#, Java, SQL, C

Machine Learning & AI: PyTorch, Scikit-learn, NumPy, Pandas, Deep Learning, Machine Learning, Neural Networks, Computer Vision, Object Detection, Object Tracking, Image Processing, Predictive Modeling, Model Training, Model Validation

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), RAG Pipelines, LLM Integration, Vector Embeddings, Semantic Search, LangChain, Ollama, Model Context Protocol (MCP), Token Optimization

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Design Patterns, Multithreading & Concurrency, Software Design, Debugging, Troubleshooting, Test Automation, Unit Testing, Integration Testing, Software Validation, SDLC

Databases: MySQL, SQL Server, PostgreSQL, Oracle, MongoDB

Tools & Methodologies: Git, Bitbucket, Jira, Streamlit, Agile, Scrum, Test-Driven Development (TDD), Code Review, Technical Documentation

PROFESSIONAL EXPERIENCE

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Project: [MiGa: Multi-Chicken Gait Assessment](#)

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking (MOT), and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a Multi-Bird Detection split of 4,569 frames with 11,422 bounding boxes and a Gait-Score Regression split spanning 55 expert-annotated sequences
- Implemented spatial normalization to map keypoints into local coordinate spaces to eliminate translational variance, achieving a keypoint mAP of 91.83% at 1,250 FPS alongside 0.93 MOTA and 0.91 IDF1 tracking accuracy

Project: [DualFit: Two-Stage Virtual Try-On via Warping and Synthesis](#)

- Developed a PyTorch-based two-stage Virtual Try-On (VTON) model combining flow-based warping and Res-UNet image synthesis
- Designed preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Achieved 24.9 PSNR, 0.913 SSIM, and 5.261 FID on the VITON-HD dataset

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation.
- Implemented reusable software components and automated workflows within an object-oriented application architecture, improving development and testing efficiency.
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements.
- Integrated automated testing and validation workflows to improve test coverage, reduce manual execution, and support repeatable software verification.
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, optimize workflows, and deliver production-ready software.

PROJECTS

Expense Tracker with Predictive Analytics

C++, MySQL, Python, SQL, Database Integration

- Designed and implemented a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data.
- Developed and optimized SQL queries to generate category-based expense summaries and reports, integrating database operations with the application's business logic.
- Integrated Python-based analytics with the application workflow to analyze historical spending patterns and generate next-month expense predictions.

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama

- Developed an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, and process unstructured web content on-demand
- Designed a local vector storage architecture using FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion steps and reducing processing latency
- Integrated a local LLM retrieval system via LangChain to orchestrate semantic queries against the vectorized data using local Llama 3.2 instances